

34.5-1 First we prove that SUBGRAPH-ISOMORPHISM is in NP. Let (G_1, G_2) be an instance. Assume the nodes of each graph are numbered $1, 2, \dots, |G|$. Let x be a witness, namely a list of nodes $i_1, i_2, \dots, i_n$ where $n = |G_1|$.

Verifying that G_1 is isomorphic to a subgraph of G_2 now means verifying that the subgraph induced by the node list x is equal to G_1 . This can be done by for each i_k, i_l verify that $(i_k, i_l) \in E_1$ if and only if $(k, l) \in E_2$. Since there are $O(n^2)$ such pairs, the check takes polynomial time. This proves that SUBGRAPH-ISOMORPHISM is in NP.

Note: It is important the witness actually is given as a node *list*, not as a set, since determining whether two graphs are isomorphic is believed to be hard in general.

We now show that SUBGRAPH-ISOMORPHISM is NP-hard. We show this by using SUBGRAPH-ISOMORPHISM to solve the known NP-complete problem CLIQUE. Let (G, k) be an instance of clique where we want to determine whether the graph G contains a clique of size k . We create the instance of SUBGRAPH-ISOMORPHISM (G, G_k) , where G_k is the complete graph with k nodes.

We show that these two instances are equal.

If G has a k -clique, then G has the complete k -graph as subgraph, and the instance (G, G_k) is also a yes-instance.

If G has G_k as subgraph, then it has a k -clique as subgraph, and hence (G, k) is a yes-instance of CLIQUE.

We conclude that SUBGRAPH-ISOMORPHISM is NP-hard and hence NP-complete.

34.5-7 We begin by verifying that LONGEST-SIMPLE-CYCLE is in NP. Let (G, k) be an instance where we want to determine whether or not there exists a cycle of length k . Let x be a list of nodes. It can be verified in polynomial time that x is indeed a cycle, that no vertex is in x more than once and that the length of x is k or greater. Hence LONGEST-SIMPLE-CYCLE is in NP.

We now show that LONGEST-SIMPLE-CYCLE is NP-hard by showing how to reduce HAMILTONIAN-CYCLE to LONGEST-SIMPLE-CYCLE. Let G be an instance of HAMILTONIAN-CYCLE. We let $(G, |G|)$ be an instance of LONGEST-SIMPLE-CYCLE. It is clear that G has a Hamiltonian cycle exactly when G has a cycle of length $|G|$. Hence LONGEST-SIMPLE-CYCLE is NP-hard.

These two results together show that LONGEST-SIMPLE-CYCLE is NP-complete.

- Show that HAMILTONIAN-PATH is NP-complete. A graph has a Hamiltonian path if there is a path that passes each node exactly once.

In NP: We can in polynomial time verify that a given path is indeed a Hamiltonian path. We leave out the details.

NP-hard: We reduce from HAMILTONIAN-CYCLE, which is known to be NP-complete. We want to construct the reduction by for a given graph $G = (V, E)$ construct a graph $G' = (V', E')$ such that G' has a Hamiltonian path exactly when G has a Hamiltonian cycle. If we can do that, we can construct an algorithm that solves HAMILTONIAN-CYCLE by using HAMILTONIAN-PATH a sub-routine. This would prove that HAMILTONIAN-PATH is at least as hard HAMILTONIAN-CYCLE (up to a polynomial factor).

The construction is as follows. Let $u \in V$ be an arbitrary vertex. We add a copy of u (together with all edges) and call it u' . Then we add two vertices v and v' and the edges (u, u') and (v, v') . We call the result G' . We now verify that G' has the properties we need. Assume G has a Hamiltonian cycle. Then G' has a path starting in u and ending in u' . We can extend this path to a Hamiltonian path by adding the edges (u, u') and (v, v') .

Now assume that G' has a Hamiltonian path. Obviously this path must have v and v' as end-points, which implies that there must be a path from u to u' . We can then close the path by letting removing the edge to u' and let it go to u instead. (From the construction of u' this is always possible.)

We have proved that the reduction is correct and that HAMILTONIAN-PATH is NP-complete.

- We now consider the problem MINIMUM-DEGREE-SPANNING-TREE: Given a graph G and an integer k , decide whether G contains a spanning tree T such that the degree of each node in T is at most k .

We show that this problem is in NP. Let (G, k) be an instance and let the witness x be a spanning tree. We need to check that x is indeed a tree and that the degree of the nodes in x don't exceed k . This can be done in polynomial time.

We now give a reduction from HAMILTONIAN-PATH, which is known NP-complete from the previous task. Let G be an instance of HAMILTONIAN-

PATH. We let $(G, 2)$ be an instance MINIMUM-DEGREE-SPANNING-TREE. We claim that G has a hamiltonian path exactly when G has a spanning tree of degree at most 2. Assume such a spanning tree exists. With degree 2 it cannot branch, and since the tree spans the graph, only two vertices can have degree 1. Therefore this tree is indeed a Hamiltonian path. On the other hand, if the graph has a Hamiltonian path, this path is obviously a spanning tree of degree 2. Therefore the reduction is correct.

We can conclude that MINIMUM-DEGREE-SPANNING-TREE is NP-complete.